

Supplemental Material

GeoFormerX: Parameter-efficient multimodal foundation-model adaptation for structured pavement condition information extraction

Reproducibility identifiers and preprocessing constants

Table S1 records the literal code, configuration, checkpoint, split, source-revision, and normalization identifiers used by the reported experiments. The values are provided in this supplemental file so that the main-manuscript claims and their exact computational provenance remain together.

Table S1. Exact reproducibility identifiers, split definitions, and full-precision preprocessing constants.

Item	Exact value	Note
GeoFormerX experiment code-tree SHA-256	3deee9a32e1aec6a2d927e2e69aea2a416a57ce40dcd3e182054f7388cdfee08	SHA-256 of the code tree recorded for all three reported GeoFormerX runs; not represented as a Git commit.
GeoFormerX canonical configuration SHA-256	de9648b4ee6316d37526af7b9e258179fcb4cc5db572bd6444e4eda2a9aa90d	SHA-256 of the canonical GeoFormerX-G8-D0-S0 configuration shared across the three reported runs.
Normalized operational protocol SHA-256	d1c12acd1f738b05e0f2b0f12488fb059fb3d7abe8d95d69dff89d6ad0724243	SHA-256 of the normalized operational protocol shared across the three runs; seed- and orchestration-only fields are excluded.
GeoFormerX selected checkpoint SHA-256s	seed 2026: b7aaa51973f2f6f65d93774f8ce3234bfaae62d33e69d7b70eda9e81f9126a36 seed 2027: c3b32e74197b0902d1cefef6a51a48f94f25930d2ca183f94ec2b6a778fcd15d seed 2028: d1faa0fd20fc4625b86913e398740ab87401743049d86e69d404774d1f1dace7	SHA-256 values of the three selected GeoFormerX checkpoints. Seed 2028 is the prespecified primary model reported in the main comparison.
SAM ViT-B checkpoint SHA-256	ec2df62732614e57411cdcf32a23ffdf28910380d03139ee0f4fcbe91eb8c912	SHA-256 of sam_vit_b_01ec64.pth used to initialize the frozen SAM backbone.
SegFormer-B2 source revision	65fa8cfa9b52b6ee7e8897a98705abf8570f9e32	Official NVlabs SegFormer Git commit used in the reported baseline experiments.
CMX source revision	e251d860aebc2f583a6c4919877e6bebe7f1aff3	Official CMX Git commit used in the reported baseline experiments.
FrozenSAM matched configuration SHA-256	457225d974830b4b242297c42cd2f295c8e35ac70d28d65bcc374949f134465a	SHA-256 of the matched FrozenSAM configuration included in the final paper evidence package.
SegFormer-B2 configuration SHA-256	b55412446810b1708879d111872766db7c55cc8cb02e57c46fff4499492bc3c7	SHA-256 of the SegFormer-B2 configuration included in the final paper evidence package.
CMX configuration SHA-256	072c84b015ae1d2e6431a79a8ad88a0275b1afab022203a5f8392c5ee629f2c5	SHA-256 of the CMX configuration included in the final paper evidence package.
Training ID-list SHA-256	fe4abc0c3272063a0503ab1408ea18e94842e4044c0f82359287b0dc86cddacd	SHA-256 of the canonical ordered train.txt file defining the 7,000-image training split.

Item	Exact value	Note
Source-validation ID-list SHA-256	0401afafa2b4f9b4d877c7296d7b70783bcea4db774dbe879c597b43f74da801	SHA-256 of the canonical ordered source_val.txt file defining the 1,000-image source-validation split.
Official test-split definition SHA-256	d0eb34de25c9d38fcf74599ee57e0e1cfaeba4d6ed692992d8bff3dcd5005d4d	Digest recorded for the canonical 1,000-ID official test-split definition used in the reported fixed-checkpoint evaluation.
CMX range mean	0.4972576734826013	Computed from the 7,000-image training set after division by 255.
CMX range population standard deviation	0.04866906799254494	Computed from the 7,000-image training set after division by 255.

Note. The main manuscript reports rounded values for readability. The hexadecimal strings above are literal Git revisions or SHA-256 digests and are not corrupted text. The training and source-validation digests identify the canonical ordered ID-list files. The test digest is the value recorded for the canonical 1,000-ID test-split definition in the final paper evidence package. The adaptation pool was not used for model fitting, architecture or checkpoint selection, or the reported fixed-checkpoint comparison; no adaptation-pool digest is reported here.

Additional deterministic qualitative examples

Figures S1–S6 present the six test IDs selected by equally spaced position in the canonical 1,000-image test list, independently of model performance. Each panel shows the grayscale-intensity raster, range-coded raster, reference mask, and predictions from GeoFormerX, matched FrozenSAM, and CMX. The examples support visual inspection of representative outputs; quantitative conclusions are based on the complete test set.

Fig. S1. Deterministic test sample DL2D00000006 (position 1/1000; reference classes: Background, Crack, and Road Marking). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX.

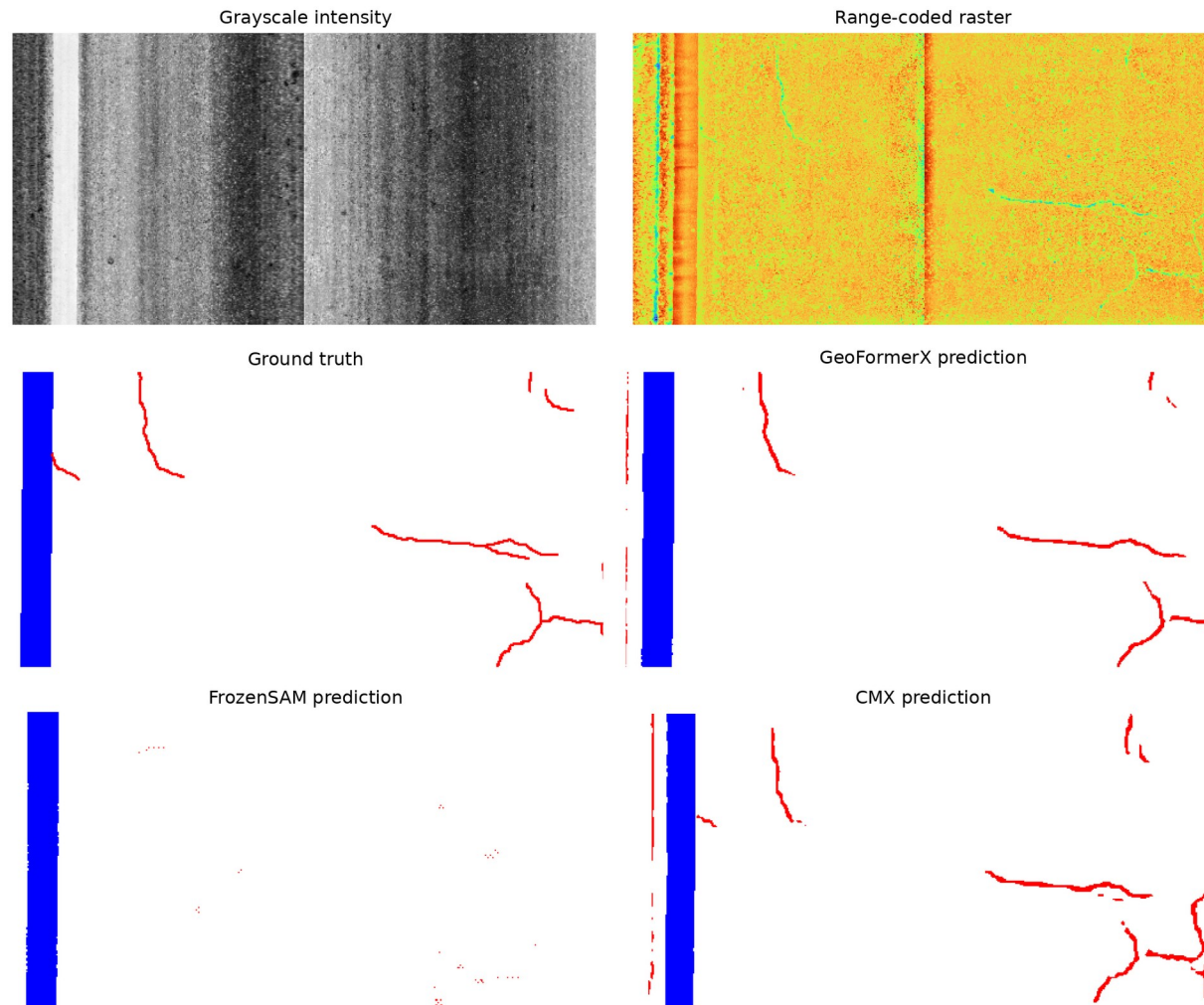


Fig. S2. Deterministic test sample DL2D00002092 (position 201/1000; reference classes: Background and Sealed Crack). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX.

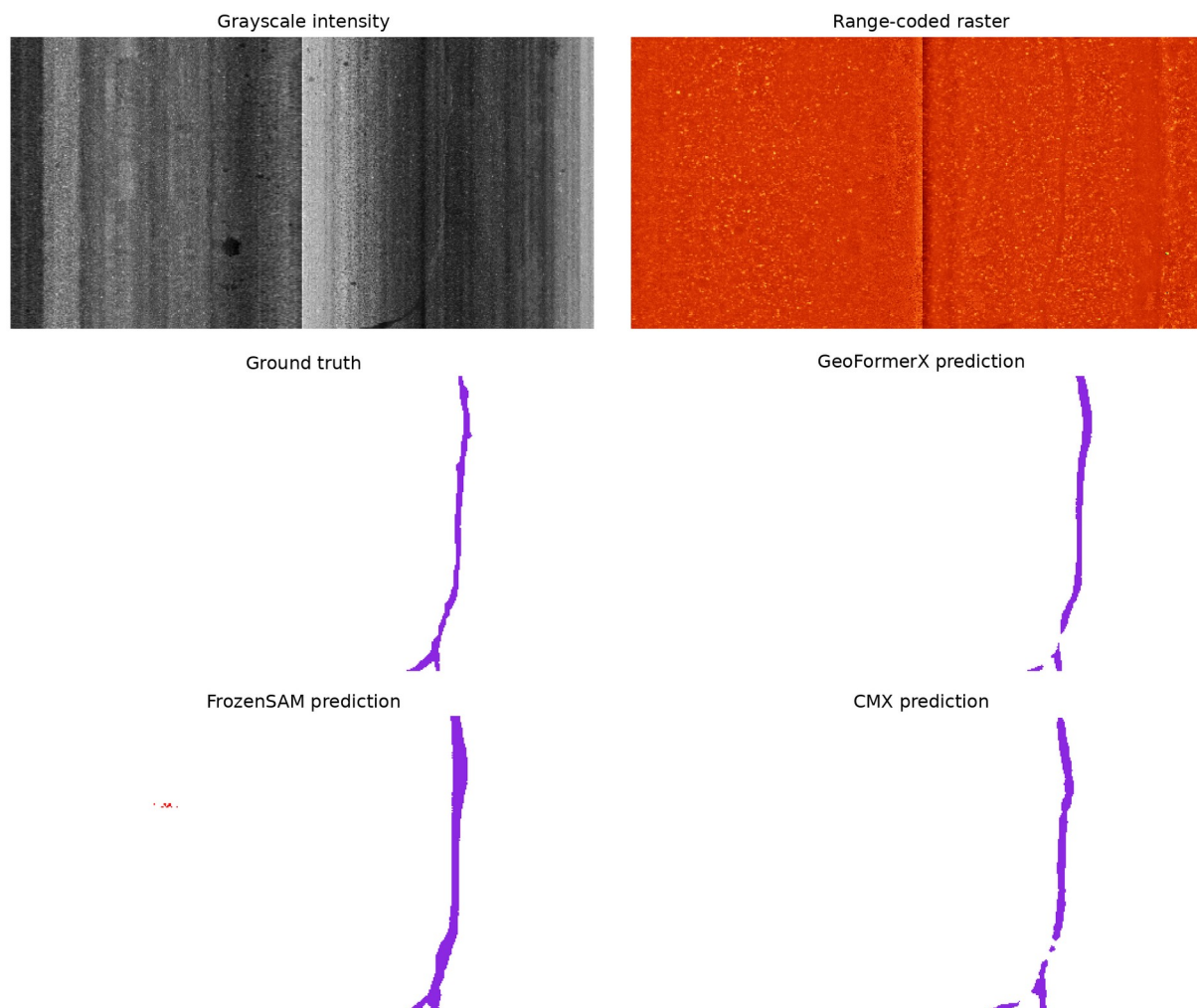


Fig. S3. Deterministic test sample DL2D00004126 (position 401/1000; reference classes: Background, Sealed Crack, and Road Marking). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX.

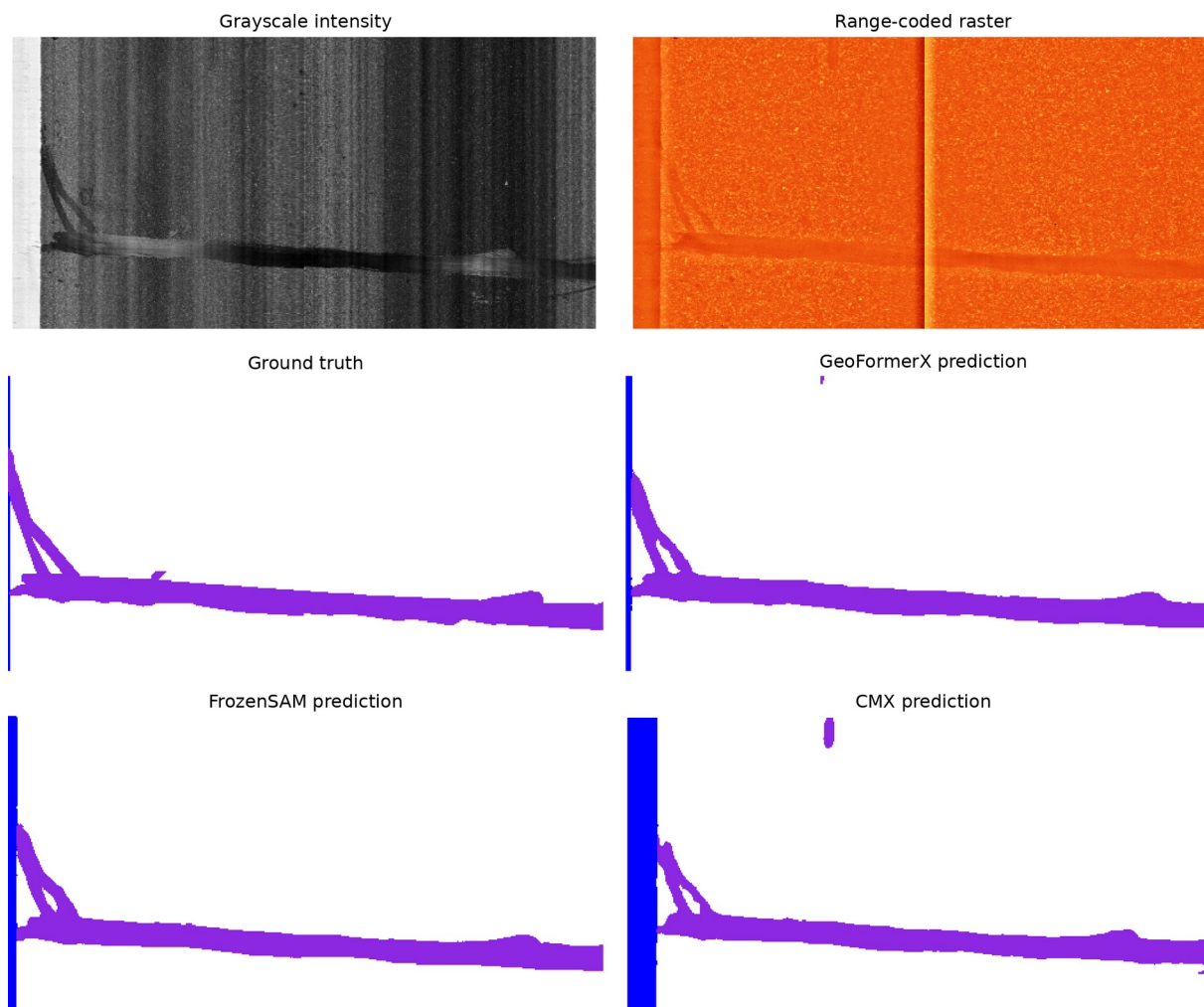


Fig. S4. Deterministic test sample DL2D00006080 (position 600/1000; reference classes: Background, Sealed Crack, and Road Marking). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX. This sample is also shown in Fig. 7 of the main manuscript.

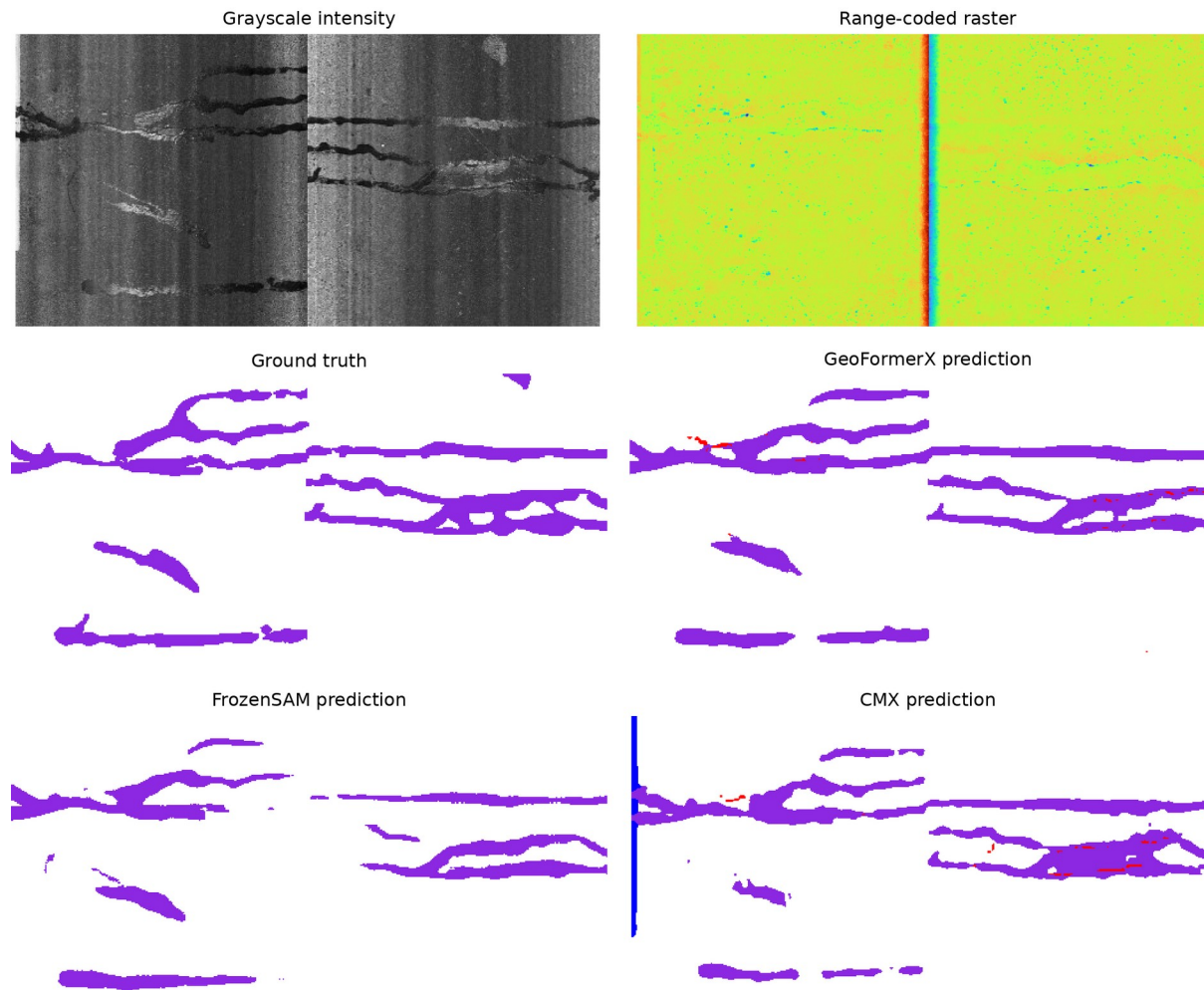


Fig. S5. Deterministic test sample DL2D00007993 (position 800/1000; reference classes: Background and Road Marking). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX.

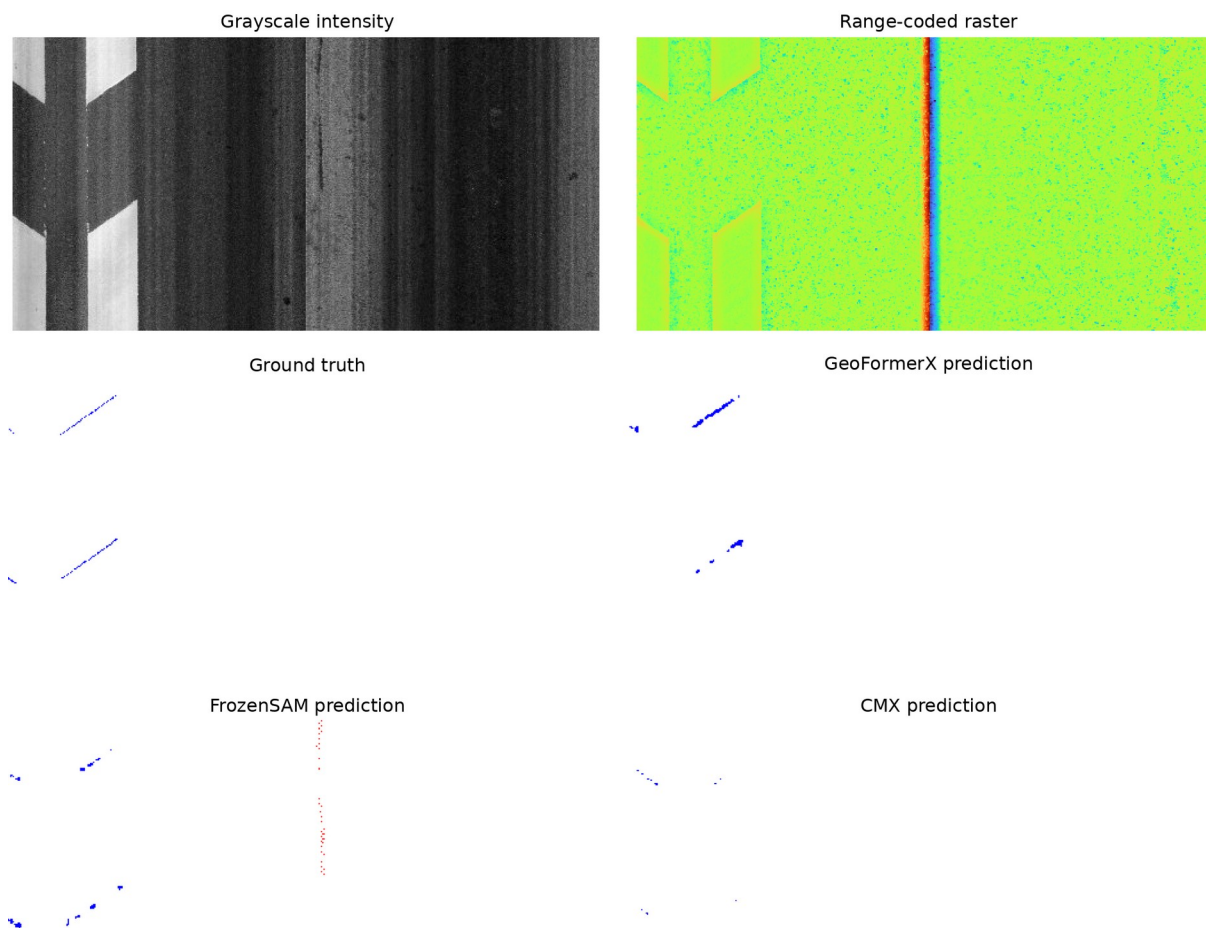


Fig. S6. Deterministic test sample DL2D00009993 (position 1000/1000; reference classes: Background and Road Marking). Panels show grayscale intensity, range-coded raster, reference mask, and predictions from GeoFormerX, FrozenSAM, and CMX.

